

Assignment 2

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Task 1

1.

The two main functions of an operating system are managing computer resources and providing an extended machine. It manages resources such as the CPU, memory, and I/O devices, while hiding hardware complexity from users and programs.

2.

The nine types of operating systems are:

- 1. Mainframe operating systems — used for large-scale data processing, such as banking transactions.*
- 2. Server operating systems — provide services to other computers over a network, such as hosting websites.*
- 3. Multiprocessor operating systems — manage multiple processors or CPU cores for parallel processing.*
- 4. Personal computer operating systems — designed for individual users and applications.*
- 5. Handheld computer operating systems — designed for smartphones and tablets.*
- 6. Embedded operating systems — used in special-purpose devices such as cars and smart devices.*
- 7. Sensor-node operating systems — used in small devices that collect and process sensor data.*
- 8. Real-time operating systems — used when tasks must be completed within strict time limits.*
- 9. Smart-card operating systems — used in small computers embedded in smart cards for identification or payments.*

3.

Multiprogramming keeps several programs in memory and switches to another program when one is waiting for I/O. Its main goal is to keep the CPU busy. Timesharing gives each process a small time slice so users can interact with the system quickly.

4.

User mode is used by application programs and has limited access to system resources. Kernel mode is used by the operating system and has full access to hardware and privileged instructions. The separation protects the operating system from faulty or malicious programs.

5.

I/O instructions are privileged because they directly control hardware devices and system resources. Allowing user programs to execute them directly could cause security problems or damage system data.

6.

Performance and reliability can conflict because additional safety checks can improve reliability but reduce performance. Resource utilization and timeliness can also conflict because efficient resource use may sometimes reduce the ability to guarantee fast and predictable responses.

Task 2

Chapter 6

The screenshot shows the Cisco Networking Academy interface for the Linux Essentials course, specifically Chapter 6. The browser address bar displays a URL from netacad.com. The page header includes the Cisco Networking Academy logo, the NDG logo, and the text "Linux Essentials". A navigation bar at the top shows "Chapter 6" with a back arrow and an "Assignment Content" link. The main content area is divided into two sections. On the left, under the heading "INSTRUCTIONS", there is a sub-heading "Chapter 6" and a green button with a right arrow and the text "Chapter 6". On the right, there is a large blue banner with the text "Linux Essentials" and "Chapter 06". Below this banner is a white box containing a green circular button with a right arrow and the text "Launch Activity". The background of the banner shows a person's hands typing on a laptop keyboard.

Chapter 6 Exam

Chapter 06 Exam

POINTS
10

ATTEMPTS
0 / Unlimited

CATEGORY
Linux Essentials

INSTRUCTIONS
Chapter 06 Exam

Chapter 06 Exam

Chapter 06 Exam

Activity Completed

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Chapter 6 Lab

Cisco
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Academy

NDG

Linux Essentials

Lab 06

POINTS
5

ATTEMPTS
0 / Unlimited

CATEGORY
Linux Essentials

INSTRUCTIONS
Lab 06

Lab 06

Lab 06

Linux Essentials

Lab 06

Launch Activity

Chapter 7 Exam

←

Chapter 07 Exam

Assignment Content

POINTS
10

ATTEMPTS
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CATEGORY
Linux Essentials

INSTRUCTIONS

Chapter 07 Exam

▶ Chapter 07 Exam

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Chapter 7 Lab

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Lab 07

Assignment Content

POINTS
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ATTEMPTS
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CATEGORY
Linux Essentials

INSTRUCTIONS

Lab 07

▶ Lab 07

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Linux Essentials

Lab 07

➔ Launch Activity

Chapter 8 exam

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Linux Essentials

 

 Chapter 08 Exam

Assignment Content

POINTS
10

ATTEMPTS
0 / Unlimited

CATEGORY
Linux Essentials

INSTRUCTIONS
Chapter 08 Exam

 Chapter 08 Exam

Chapter 08 Exam

Activity Completed

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Chapter 8 Lab

 Lab 08

Assignment Content

POINTS
5

ATTEMPTS
0 / Unlimited

CATEGORY
Linux Essentials

INSTRUCTIONS
Lab 08

 Lab 08

Lab 08

Linux Essentials

Lab 08

 Launch Activity